

Intelligent AI Delegation: Framework Mapping

A detailed comparison of Google DeepMind's theoretical framework and DIR/ROA's production implementation

Overview

On February 12, 2026, Google DeepMind published ["Intelligent AI Delegation"](#) (arXiv:2602.11865). Reading it revealed remarkable convergence with Responsibility-Oriented Agents (ROA) and Decision Intelligence Runtime (DIR) - architectural patterns I've been developing since 2025 based on production experience with Alvestor, an autonomous trading system.

This document maps the two frameworks, showing how independent efforts converged on the same architectural primitives when solving the "Day Two" problem of reliable AI systems.

Problem Statement Alignment

Dimension	Google DeepMind	DIR/ROA
Core Issue	<i>"Existing task decomposition and delegation methods rely on simple heuristics, and are not able to dynamically adapt to environmental changes and robustly handle unexpected failures." (§1)</i>	<i>"Most current 'agent' frameworks wrap LLM calls in orchestration layers, hoping that structure will translate into coherent behavior. In practice, these systems often lack determinism, boundaries, accountability..." (ROA §0)</i>
Root Cause	<i>"Real-world AI deployments need to move beyond ad hoc, brittle, and untrustworthy delegation." (§1)</i>	<i>"Agents today do not have responsibility - a defining property of any real decision-making unit." (ROA §0)</i>
Architectural Gap	Task decomposition without authority transfer, accountability, or	Probabilistic reasoning directly coupled to deterministic execution without

Dimension	Google DeepMind	DIR/ROA
	trust mechanisms	governance layer

References:

- Google: §1 (Introduction)
- ROA: Abstract (§0), §2.1–2.2
- DIR: §1 (Motivation)

Core Concept Mapping

1. Delegation as Responsibility Transfer

Aspect	Google DeepMind	DIR/ROA
Definition	<i>"Intelligent delegation [is] a sequence of decisions involving task allocation, that also incorporates transfer of authority, responsibility, accountability, clear specifications regarding roles and boundaries, clarity of intent, and mechanisms for establishing trust."</i> (§2.1)	<i>"Each agent in ROA is defined not by its personality or its toolset, but by a Responsibility Contract - a formal description of its role in the decision architecture."</i> (ROA §3.1)
Components	Authority, responsibility, accountability, roles, boundaries, intent, trust (§2.1)	Scope, authority, mission, boundaries, escalation criteria (ROA §3.1)
Implementation	Delegation protocols with role specifications (§4.7)	Agent Registry as single source of truth; no self-granted permissions (DIR §2.3, §6.2)

Key Quote (ROA):

"Responsibility is the only abstraction that simultaneously binds authority, memory, escalation, and accountability into a single, governable unit." (ROA §1)

References:

- Google: §2.1 (Definition), §4.7 (Permission Handling Protocol)
- ROA: §3.1 (Responsibility Contract), §3.3 (Mission)
- DIR: §2.3 (Agent Registry)

2. Auditability and Traceability

Aspect	Google DeepMind	DIR/ROA
Requirement	<i>"Strictly enforced auditability with attribution for both successful and failed executions"</i> (§4, §4.5, §4.8)	<i>"Every Policy is treated as a Claim until validated against schema, RBAC rules, and state consistency."</i> (DIR §5.3, §6)
Mechanism	Delegation chains with attestation (§4.5); verifiable task execution via zero-knowledge proofs, smart contracts (§4.8)	DecisionFlow ID (DFID) - correlation identifier binding: trigger → context → reasoning → proposal → validation → execution (DIR §4, §6)
Scope	Transitive accountability in delegation networks (A→B→C) (§4.5)	Parent-child DFID relationships; Saga patterns for distributed transactions (DIR Topologies §2.4)

Key Quote (DIR):

"In a microservice, we use a Trace ID (OpenTelemetry) to follow a request. In an agent system, we need to trace the reasoning chain that led to an action." (DIR §1.2)

References:

- Google: §4 (Framework), §4.5 (Monitoring - Agent Attestation), §4.8 (Verifiable Task Completion)
- DIR: §4 (DecisionFlow ID), §5.3 (Claims vs. Facts), §6 (Decision Integrity Module)
- DIR Topologies: §2.4 (Attestation in EOAM)

3. Permission Handling and Authority Boundaries

Aspect	Google DeepMind	DIR/ROA
Principle	<i>"Strict roles and bounded operational scopes with</i>	<i>"Agents cannot self-grant permissions. The separation between Kernel Space</i>

Aspect	Google DeepMind	DIR/ROA
	<i>sufficient privileges without excessive risk" (§4.7)</i>	<i>(deterministic execution) and User Space (probabilistic reasoning) is absolute."</i> (DIR §2.3)
Architecture	Permission Handling Protocol ensuring delegates operate within granted authority (§4.7)	Kernel Space vs. User Space: Agents propose in User Space; Runtime validates and executes in Kernel Space (DIR §2)
Enforcement	Role-based access control; capability matching (§4.2, §4.7)	Decision Integrity Module (DIM) as Policy Enforcement Point; schema validation, RBAC, risk checks (DIR §6, §6.2)

Key Quote (DIR):

"Prompts are not permissions... Safety mechanisms become prompts ('Please do not trade if volatility is high') rather than hard constraints. As any security engineer knows, prompts are not permissions." (DIR §1.1)

References:

- Google: §4.2 (Task Assignment), §4.7 (Permission Handling)
- DIR: §2 (Kernel vs. User Space), §6 (Decision Integrity Module), §6.2 (Validation Pipeline)

4. Dynamic Cognitive Friction and Escalation

Aspect	Google DeepMind	DIR/ROA
Concept	<i>"Zone of indifference" - delegates may execute without scrutiny, allowing intent mismatches to propagate. Requires "dynamic cognitive friction" - agents that challenge or request verification when appropriate (§2.3)</i>	<i>"When a request is contextually ambiguous, the agent 'steps outside' rather than proceeding blindly."</i> via Self-Check logic and explicit escalation criteria (ROA §4, §5.3)
Mechanism	Switching delegates mid-execution; re-delegating on failure or context shift (§4, §4.4)	Soft-stops, hard-stops, delegation to higher-level agents; Priority-Based Preemption in EOAM

Aspect	Google DeepMind	DIR/ROA
		topology (DIR §9; DIR Topologies - EOAM)
Trust Calibration	Aligning trust levels with true capabilities, supported by explainability (§2.3)	Agents as epistemic entities : <i>"They interpret and propose but provide no safety or enforcement guarantees."</i> Explain (narrative) + Policy (structured intent) (ROA §4.1–4.4)

Key Quote (ROA):

"ROA agents are epistemic entities. They interpret, explain, and propose. They provide no safety, correctness, or enforcement guarantees." (ROA §0)

References:

- Google: §2.3 (Delegation in Human Organizations), §4.4 (Adaptive Coordination)
- ROA: §4 (Decision Lifecycle), §4.1–4.4 (Explain, Policy, Self-Check, Proposal), §5.3 (Escalation)
- DIR: §9 (Human-in-the-Loop)

5. Task Characteristics and Execution Parametrization

Aspect	Google DeepMind	DIR/ROA
Task Dimensions	Criticality, reversibility, verifiability, contextuality, complexity, uncertainty (§2.2)	Execution parametrization: TTL (Time-to-Live), drift envelopes, idempotency requirements (DIR §3.3, §6.4, §6.5)
Operational Impact	Task characteristics inform delegation strategy and risk assessment (§2.2)	Decision Integrity Module + Escalation Manager handle tasks according to characteristics; expired decisions (TTL breach) are discarded, not delayed (DIR §1.2, §6.4)
Verification	Contracts ensuring execution faithfully follows request;	Every Policy treated as Claim until validated against schema, RBAC, state consistency (DIR §5.3, §6)

Aspect	Google DeepMind	DIR/ROA
	formal verification mechanisms (§4.2, §4.8)	

Key Quote (DIR):

"Data expires. An agent's intent must have a strict validity window. If the runtime cannot execute the decision within that window, it must be discarded, not delayed." (DIR §1.2)

References:

- Google: §2.2 (Aspects of Delegation), §4.2 (Task Assignment), §4.8 (Verifiable Task Completion)
- DIR: §1.2 (Time-to-Live), §3.3 (Execution Parametrization), §6 (DIM), §6.4 (TTL), §6.5 (JIT State Verification)

Architectural Pluralism: No Universal Pattern

Both frameworks reject monolithic solutions, advocating for **contingency-based design**.

Dimension	Google DeepMind	DIR/ROA
Theoretical Basis	<i>"Contingency theory: no universally optimal structure exists; design must be dynamically matched to the task"</i> (§2.3)	<i>"A single architectural pattern cannot satisfy conflicting requirements."</i> (DIR Topologies §1)
Delegation Types	Atomic vs. open-ended delegation (§2.1, §4.2); decentralized market hubs, auction queues (§4.2, §4.4); zero-knowledge proofs, smart contracts (§4.8)	Three topologies by decision class: EOAM (mesh), SDS (stream), DL+PCI (ledger) (DIR Topologies §1–4)
EOAM/Mesh	Decentralized coordination via market mechanisms, hubs (§4.2, §4.4)	Event-Oriented Agent Mesh: decentralized choreography through event substrate; Priority-Based Preemption for complex strategy (DIR Topologies §2)

Dimension	Google DeepMind	DIR/ROA
SDS/Stream	Atomic task delegation with bounded scope (§2.1, §4.2)	Sovereign Decision Stream: high-velocity atomic execution; constrained decoding ensures outputs are "syntactically bound by design" (DIR Topologies §3)
DL+PCI/Ledger	Verifiable completion via zero-knowledge proofs, smart contracts (§4.8)	Decision Ledger with Proof-Carrying Intents: formal verification, zero-trust execution based on verified artifacts (DIR Topologies §4)

References:

- Google: §2.1 (Definition), §2.3 (Delegation in Human Organizations), §4.2 (Task Assignment), §4.4 (Adaptive Coordination), §4.8 (Verifiable Task Completion)
- DIR Topologies: §1 (Introduction), §2 (EOAM), §3 (SDS), §4 (DL+PCI)

Registries and Accountability Infrastructure

Component	Google DeepMind	DIR/ROA
Agent Registry	<i>"Registries of agents, tools, and human participants that list skills, past activity, completion rates, and availability"</i> (§4.2)	Agent Registry as single source of truth: capability manifests, lifecycle management, schema versioning (DIR §2.3; ROA §6.4)
Delegation Chains	Transitive accountability via attestation for chains $A \rightarrow B \rightarrow C$ (§4.5)	Parent-child DFID relationships; Saga patterns; signed attestations at each agent boundary in EOAM (DIR §4.2; DIR Topologies §2.4)
Capability Matching	Skill-based discovery and capability lookup (§4.2)	Schema-based capability manifests; version compatibility checks (DIR §2.3)

References:

- Google: §4.2 (Task Assignment), §4.5 (Monitoring)

- DIR: §2.3 (Agent Registry), §4 (DecisionFlow ID)
- ROA: §6.4 (Agent Registry)
- DIR Topologies: §2.4 (Attestation in EOAM)

Theory vs. Implementation

Where Google Explores

Area	Description
Market Mechanisms	Reputation systems, auction-based task allocation, decentralized agent economies (§4.2, §4.4, §4.6)
Cryptographic Verification	Zero-knowledge proofs, smart contracts at scale for verifiable completion (§4.8)
Multi-Objective Optimization	Balancing efficiency, safety, and human welfare across delegation networks (§2.3, §4)
Human-AI Interaction	AI-to-human delegation, human welfare considerations, moral decision-making boundaries (§2.1, §2.3)

Where DIR/ROA Implements

Area	Description
Concrete Patterns	DIM, Agent Registry, DFID, idempotency, TTL, Saga - production-ready code (DIR §2–§9)
Engineering Primitive	Kernel vs. User Space separation as foundational architectural constraint (DIR §2)
Topology Blueprints	Three opinionated topologies with architecture diagrams and reference implementations (DIR Topologies)
Public Stress-Test	Alvestor (2025–2026) as dated, documented real-world validation (ROA §1; DIR §1)

Summary:

- **Google:** Theoretical framework for the emerging agentic web
- **DIR/ROA:** Blueprint for teams building production systems today

Convergence Significance

When independent efforts—one from a research lab exploring delegation theory, one from production system stress-testing—converge on the same architectural primitives, it signals an inflection point.

Shared Primitives:

1. **Responsibility** as the primary abstraction (not capability or role)
2. **Boundaries** as hard constraints (not prompt suggestions)
3. **Auditability** through structured tracing (not post-hoc logs)
4. **Separation** of reasoning and execution (not interleaved loops)
5. **Pluralism** in delegation patterns (not universal frameworks)

The shift:

- **From:** "Wrap an LLM in a loop and call it an agent"
- **To:** "Agents as bounded units of responsibility, runtimes as policy enforcement points, decision lifecycles that can be traced, validated, and trusted"

References

Google DeepMind:

- Tomašev, N., Franklin, M., Osindero, S. (2026). *Intelligent AI Delegation*. Google DeepMind. arXiv:2602.11865v1 [[cs.AI](#)]. 12 Feb 2026.
 - <https://arxiv.org/abs/2602.11865>
 - All "§X" citations reference sections in this paper

DIR/ROA:

- [ROA Manifesto](#) - Responsibility-Oriented Agents architectural pattern
- [DIR Architectural Pattern](#) - Decision Intelligence Runtime implementation
- [DIR Topologies](#) - EOAM, SDS, DL+PCI operational modes
- Repository: <https://github.com/huka81/decision-intelligence-runtime>

Discussion

Are you building production AI systems? What "Day Two" failures have you encountered? How are you solving the responsibility and execution problem?

The convergence documented here suggests the industry is moving toward governed agency as a necessity, not an option.

Tags: #AIAgents #DecisionIntelligence #ResponsibleAI #SystemsEngineering #AIArchitecture